ScholAR: Page-Grounded Retrieval-Augmented Generation for Scientific Document Comprehension

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Abstract

Understanding a scientific paper means being able to trace each claim back to the exact place it came from. General-purpose chat over an uploaded PDF is fluent but unreliable for this: nothing ties a cited page number to the text the model was actually shown, and the scientific retrieval-augmented systems that do better tend to rely on cloud-hosted frontier models or datastores of millions of papers. We present ScholAR (Smart Companion for Holistic Organization, Literature Analysis & Research), a retrieval-augmented generation system for scientific documents that runs entirely on a consumer laptop through a local model server, with no cloud API for text or vision. Two design choices carry the system. Page-preserving chunking keeps every retrieved unit inside a single PDF page, so a citation resolves to one highlightable location. Indirect citation grounding lets the model cite only evidence identifiers that map to chunks it actually retrieved, so it cannot write a page number of its own. We evaluate retrieval, answer faithfulness, visual grounding, and cross-document localization on labeled benchmarks. On 51 cases, the generated answers stay grounded in their retrieved context with a mean score of 0.937, no answer atom contradicts its source, and 92% of the 269 inline citations are fully supported with none unsupported. BM25-primary retrieval reaches Recall@5 of 0.93 across 25 papers, and figure and table questions route to the correct visual element in all 18 pilot cases. On M3SciQA's cross-document localization task, retrieving from the question alone sits near chance, but resolving the anchor figure with the same local multimodal model first reaches an MRR of 0.474, ahead of every open-source baseline reported there and close to GPT-4o (0.500), while running on one machine. Against local reimplementations of PDF-chat, vanilla RAG, and a PaperQA2-style pipeline, ScholAR leads on answer correctness and faithfulness, and when a question's answer is absent from the retrieved paper the local models decline rather than fabricate on 90 to 100% of a provably-unanswerable set, each within an 18 GB single-machine budget. We also contribute a 100-case benchmark across 25 papers spanning text, mathematical, and figure questions, and a model-agnostic human-evaluation protocol over four local models whose expert scoring is pending.

1 Introduction

Reading an unfamiliar paper is slowest at the moments that matter most: checking that a stated number, a method detail, or a figure actually says what you think it says. Uploading the PDF to a general-purpose chat model helps with fluency but brings a new failure mode: such systems summarize confidently while inventing page references and attributing quotes to places they do not appear, because a generic chat pipeline puts no hard constraint between a citation and the text the model was conditioned on.

Purpose-built scientific RAG systems close part of this gap but assume resources most readers do not have. OpenScholar (Asai et al. 2024), PaperQA2 (Skarlinski et al. 2024), and SciRAG (Ding et al. 2025) depend on cloud-hosted frontier models, and OpenScholar additionally on a datastore of tens of millions of papers. We ask a narrower question: what does grounded, citable question answering over a scientific document look like under a hard local-only constraint, with a single sub-15-billion-parameter model, no cloud API, and a retrieval design simple enough to audit end to end?

Two ideas carry ScholAR. Page-preserving chunking segments a document so that every retrieved chunk lies entirely within one PDF page, making a citation resolve to an exact, highlightable location. Indirect citation grounding assigns each retrieved passage a short-lived evidence identifier and instructs the model to cite only those identifiers, never a free-form page number; the application then rewrites each identifier into a numbered reference that maps deterministically to a page. The model decides only which evidence supports a claim, and the evidence-to-page mapping is application logic rather than a model guess (Figure 2). Beyond single-document text, ScholAR routes figure and table questions to a locally run multimodal model and extends to a paper's own cited references for cross-document questions.

Our contributions are:

- A fully local RAG system for scientific documents that runs both text and vision inference on-device, using page-preserving chunking and an indirect citation mechanism that rules out page-number hallucination by construction.
- An evaluation that scores the faithfulness of the system's *generated answers* rather than of pre-written gold claims, paired with an automated per-citation support metric. On 51 cases, generated answers reach a mean grounding score

of 0.937 with a zero contradiction rate and 92% fully supported citations, none unsupported.

- Automated benchmarks and results for retrieval, retrieval-support faithfulness, visual grounding, and cross-document localization, the last measured against the published baselines of M3SciQA on its own data, where a locally resolved figure lifts localization from near-chance to competitive with a frontier cloud model.
- A 100-case benchmark across 25 papers spanning text, mathematical, and figure questions, with gold answers verified against their source passages, and a model-agnostic human-evaluation protocol over four local models. The expert scoring is under way, and its result tables are marked as placeholders here.

2 Related Work

Retrieval-augmented generation. RAG conditions generation on retrieved evidence (Lewis et al. 2020). Lexical retrieval via BM25 (Robertson and Zaragoza 2009) remains a strong baseline, and BEIR (Thakur et al. 2021) documents that zero-shot dense retrievers often fail to beat it, which is why ScholAR keeps BM25 primary and treats dense retrieval as a fusion component.

Scientific-literature RAG systems. PaperQA and PaperQA2 (Lála et al. 2023; Skarlinski et al. 2024) are agentic systems reporting high accuracy with frontier backends; OpenScholar (Asai et al. 2024) synthesizes over a large open datastore, and SciRAG (Ding et al. 2025) adds citation-graph-aware, outline-guided synthesis. All three assume cloud-hosted or large self-hosted models. ScholAR instead targets a single local sub-15B model on consumer hardware, trading raw capability for zero data exposure. Closest to our citation mechanism, Pleias-RAG (Pleias 2025) *trains* small models to emit literal source quotes; we take the complementary route of putting the grounding in the retrieval and application layer, so it works with any local model rather than a model taught to quote.

Citation and faithfulness evaluation. Gao et al. (2023) formalize citation precision and recall via entailment between cited passages and statements. SummaC (Laban et al. 2022) scores consistency with zero-shot sentence-level entailment, FActScore (Min et al. 2023) decomposes text into atomic facts, and RAGAS (Es et al. 2024) automates RAG faithfulness scoring. Our metric combines SummaC-style zero-shot entailment with FActScore-style atomic decomposition, applied to the system's own generated answers rather than to gold claims.

Visual and multi-document scientific QA. ColPali (Faysse et al. 2024) indexes pages as visual embeddings without OCR; ScholAR keeps text retrieval primary and routes only figure and table questions to a cropped-region vision call. M3SciQA (Li et al. 2024) pairs each anchor paper with its cited references and finds that even large foundation models trail human experts at cross-paper reasoning; we evaluate directly on it (Section 5.5). GROBID (Lopez 2009) and Nougat (Blecher et al. 2023) target structured extraction and are complementary to our chunking.

3 Problem Formulation

Given a scientific PDF D, a user question q, and a set of extracted chunks $\mathcal{C} = \{c_1, \dots, c_n\}$ derived from D (and, in multi-document mode, from D's cited references), the system retrieves a ranked subset $\mathcal{C}_q \subseteq \mathcal{C}$ and generates an answer $\hat{a}$ grounded in that evidence:

$$\hat{a} = \arg\max_a P(a \mid q, \mathcal{C}_q), \quad \mathcal{C}_q = \mathrm{Retrieve}(q, \mathcal{C}, k). \quad (1)$$

Each chunk c_i carries a source page number and, in multi-document mode, a source paper identifier, so $\mathcal{C}_q$ always resolves to a specific page of a specific document. Generation attaches each claim in $\hat{a}$ to an evidence identifier referencing a specific $c_i \in \mathcal{C}_q$ rather than a free-form page number, so citations are constrained to what was actually retrieved, independent of $\mathrm{Retrieve}$'s own precision.

4 Method

Figure 1 shows the end-to-end pipeline. A PDF is chunked one page at a time, ranked by BM25, and exposed to the model as evidence identifiers; the model's identifier citations are then rewritten into page-anchored references by the application rather than by the model.

4.1 Document Processing and Page-Preserving Chunking

Given an arXiv identifier or an uploaded PDF, ScholAR extracts per-page text with PyMuPDF (Artifex Software 2024) and persists metadata, pages, chunks, and figures as JSON. Standard character or token windows routinely split a chunk across a page boundary, breaking the one-to-one map between a retrieved chunk and the page a citation should highlight. ScholAR instead chunks strictly within each page: it slides a window of target size 1400 words with 120-word overlap over each page's tokens independently, so a chunk never spans two pages. A lightweight regular-expression pass tags each chunk with a detected section title when a heading is found near the top of the page. Chunk identifiers are numbered per paper.

4.2 BM25-Primary Retrieval

The production retriever ranks chunks with Okapi BM25 (Robertson and Zaragoza 2009) (k_1=1.4, b=0.72) as the primary signal, then applies small additive heuristic boosts: a query-expansion dictionary maps research terms to related vocabulary, an explicit page number in the query adds a page-hint boost, and query language implying a section or a claim-bearing phrase adds a small boost. A separate hybrid configuration fuses BM25 with dense retrieval over a local all-MiniLM-L6-v2 sentence encoder (Reimers and Gurevych 2019) using Reciprocal Rank Fusion (Cormack, Clarke, and Buettcher 2009) (k=60); this configuration is used in the faithfulness study.

4.3 Indirect Citation Grounding

Asking a model to write a page number directly (for example "[p. 7]") is fragile, because the model has no reliable

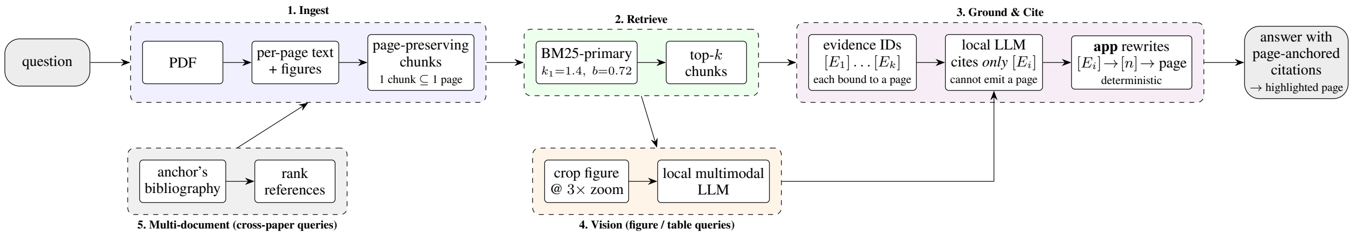

Figure 1: The ScholAR architecture. A question enters at the left and leaves as an answer whose every citation resolves to a highlightable page. **(1)** Ingest keeps each chunk inside a single PDF page. **(2)** BM25-primary retrieval returns the top-k chunks. **(3)** Those chunks are exposed to the model only as evidence identifiers $[E_i]$, each already bound to the page of the chunk it came from; the model may cite an identifier but can never write a page number, and the *application*, not the model, rewrites $[E_i]$ into a numbered reference and resolves it to a page. A cited page is therefore correct by construction rather than by the model's good behavior. **(4)** Figure and table queries additionally route a cropped, $3\times$-zoomed image to the same local multimodal model. **(5)** Cross-paper queries first rank the anchor's bibliography, and the selected reference is then ingested like any other paper. Every component runs on one machine.

way to know which page a claim came from and will produce a plausible-looking one. ScholAR instead assigns each retrieved passage a short-lived evidence identifier for the current turn and instructs the model to cite only these identifiers. After generation, the system discards any page number the model wrote on its own, rewrites each valid evidence identifier into a compact numbered reference in citation order, and resolves each reference to its source chunk's page for highlighting. The model decides only which evidence supports a claim; the mapping from evidence to page is deterministic application logic. This prevents page-number hallucination by construction rather than by asking the model to be careful.

4.4 Visual Grounding

When a query references visual content, through cue terms such as "figure" or an explicit figure number, retrieval boosts figure and table chunks so they can compete against longer text chunks. Captions are detected by regular expression and the region above each is rendered at $3\times$ zoom and cached. A query naming an explicit figure number is matched against that figure's label directly, which matters because a letter-initial tokenizer collapses "Figure 2" and "Figure 14" to the same token once digits are dropped. When the top-ranked chunk is a figure or table, the cached image is sent with its caption to the same local model used for text (both of our multimodal models read it natively), never to a cloud API; if the image is missing, the system degrades to a caption-only answer.

4.5 Multi-Document Extension

ScholAR resolves an anchor paper's bibliography and lets the user ingest cited works into the same session. For arXiv papers, references are resolved by arXiv identifier; for uploaded PDFs, by a title search over the last pages, matched against numbered and author-year citation styles. Ingesting a reference applies the same page-preserving chunking and tags every chunk with a source paper identifier. At query time, chunks from all loaded papers merge into one retrieval pool, and each citation is labeled with its source paper.

4.6 Faithfulness Metrics

We evaluate faithfulness with a zero-shot, sentence-level entailment approach in the SummaC style (Laban et al. 2022), decomposing text into atomic claims in the FActScore style (Min et al. 2023). For an atom a_i and a chunk $\mathcal{C}_j$ we take the maximum cosine similarity between the atom and the chunk's sentences under the local all-MiniLM-L6-v2 encoder, and label the atom entailment, neutral, or contradiction against thresholds τ_e=0.42 and τ_n=0.25. A Combined Faithfulness Score (CFS) blends three tiers:

$$\mathrm{CFS} = 0.50\,\mathrm{NLI} + 0.30\,\mathrm{SCR} + 0.20\,\mathrm{KFP}, \qquad (2)$$

where NLI is the entailment fraction, SCR is whole-claim to best-chunk cosine similarity, and KFP is boundary-matched recall of numbers and technical terms. The thresholds and the CFS$\geq$0.55 faithful cutoff were calibrated on a small pilot set; we did not run a systematic sensitivity sweep.

We apply this metric in two ways that answer different questions. *Retrieval-support CFS* scores a pre-written gold claim against the retrieved chunks, measuring whether retrieval surfaces support for a known fact (answerability). *Generation faithfulness* scores the system's *actual generated answer* against its retrieved context, measuring whether the output stays grounded and avoids contradicting its source. The second is the quantity our hallucination claim rests on, and it is the result we foreground.

4.7 System Architecture

The backend is a local application that stores parsed PDFs as JSON on disk; the frontend highlights cited pages. All inference, both text generation and figure or table question answering, runs locally through a single natively multimodal model, with no cloud API used for inference and no user data leaving the machine. Paper acquisition and reference resolution use public arXiv and Semantic Scholar lookups by identifier only. All experiments in this paper were run on a consumer laptop (Apple Silicon, 18 GB unified memory) under Python 3.12, with all-MiniLM-L6-v2 loaded from a local cache; no discrete or cloud GPU was used, and inference

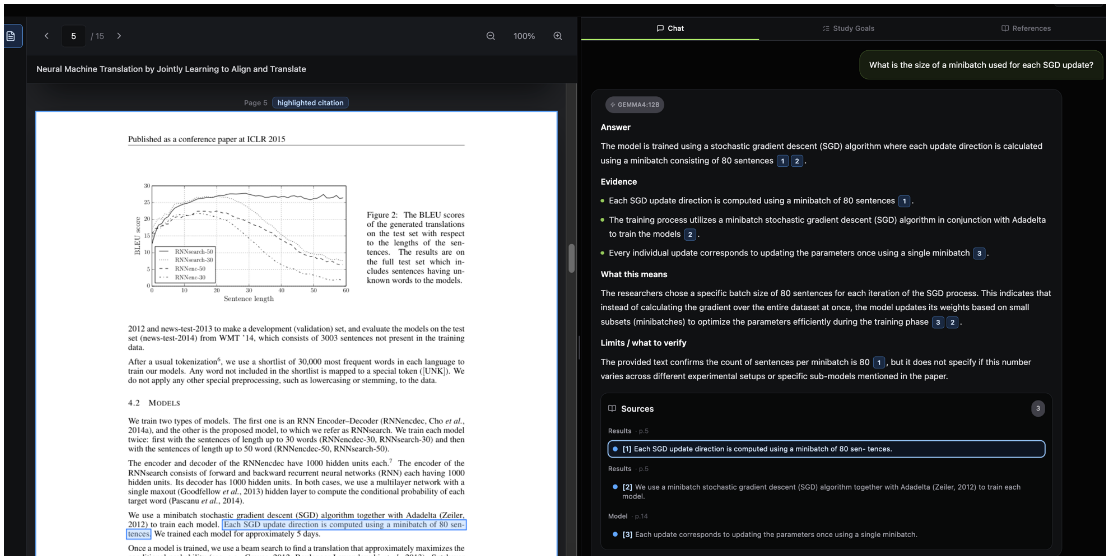

Figure 2: Indirect citation grounding as the reader sees it, captured from the running system. The question (top right) is answered by a local model (gemma4:12b) whose prose carries only numbered references, never a page number: the model emitted evidence identifiers and the application rewrote them into [1] and [2], resolving each to a page. Selecting [1] (highlighted under Sources) scrolls the document to page 5 and boxes the exact sentence the claim rests on, "Each SGD update direction is computed using a minibatch of 80 sentences." The cited page is a lookup rather than a generation, so the highlight cannot disagree with the citation. Everything shown runs on one laptop.

runs on the laptop's integrated GPU through the local model server's accelerated backend.

5 Benchmarks

We use two families of evaluation data. The first is a set of small, manually labeled benchmarks over three landmark NLP papers (Attention Is All You Need (Vaswani et al. 2017), Retrieval-Augmented Generation (Lewis et al. 2020), and LLaMA (Touvron et al. 2023)): a 14-case retrieval benchmark with labeled relevant chunks, a 51-case faithfulness benchmark with oracle claims and a supporting chunk, an 18-case visual-grounding pilot, and an 18-case multi-document benchmark of which 10 have an arXiv-resolvable target. These support the automated metrics that need ground-truth chunk labels, and they are small: the hand-labeled benchmarks cover three papers between them.

The second is a 100-case benchmark spanning 25 papers, built to broaden coverage and reduce the memorization advantage a model has on very famous papers. It contains 50 text, 25 mathematical, and 25 figure or table questions. Each case was mined from a real passage or caption by a local model and then kept only if the required answer facts appear, on a word or number boundary, in the source passage; questions from benchmark-dataset papers, reference lists, and worked examples were filtered out. Because each case carries exact substrings copied from its source passage, we recover a gold supporting chunk for every case at no extra labeling cost: it is the chunk that contains those substrings, disambiguated by the recorded page. This yields a 100-case, 25-paper labeled set for retrieval and retrieval-support faithfulness, an order of magnitude larger and eight times more papers than the hand-labeled set. The labels are weaker than hand annotation, and the questions, being mined from passages, carry lexical overlap that can favor lexical retrieval; we therefore report the scaled set alongside the 3-paper hand-labeled set rather than in place of it. This diverse benchmark is also the dataset for the human evaluation (Section 6). Table 1 summarizes the corpus and the benchmarks built on it.

Table 1: Key statistics of the ScholAR corpus and benchmarks. Page-preserving chunking yields close to one chunk per page, because a typical page holds fewer words than the chunk window.

Corpus		*Diverse benchmark*	
Papers	25	Questions	100
Pages / paper	34.5	text/math/vis.	50/25/25
Chunks / paper	34.8	Question words	12.9
Words / paper	17,694	Gold facts / q.	2.8
Labeled sets (hand / auto)		*Generated answers*	
Retrieval	14 / 100	Answers	350
Faithfulness	51 / 100	Local models	4
Visual / multi-doc	18 / 18	Inline citations	829
Unanswerable	20	Citations / answer	2.4

5.1 Retrieval-Support Faithfulness

Table 3 reports the retrieval-support CFS on the scaled 100-case set. BM25-primary and the hybrid BM25+Dense+RRF retriever land within 0.003 of each other (Combined CFS 0.785 versus 0.782), and BM25-primary has the higher

Table 2: Retrieval on the scaled benchmark: 100 auto-labeled cases across 25 papers, with NDCG@5 over the gold chunks. On the 3-paper hand-labeled anchor (14 cases) dense-only instead leads (R@5 1.000, MRR 0.881).

System	R@1	R@3	R@5	MRR	NDCG@5
Keyword	0.670	0.860	0.950	0.779	0.762
BM25-Only	0.810	0.910	0.940	0.863	0.828
BM25+Rerank	0.810	0.910	0.930	0.861	0.824
Dense-Only	0.470	0.650	0.740	0.572	0.523

Table 3: Retrieval-support faithfulness on the scaled 100-case set (25 papers). NLI is the entailment fraction; SCR is whole-claim cosine; KFP is key-fact recall; SCHR@5 is the supporting-chunk hit rate at depth 5. On the 3-paper anchor (51 oracle-claim cases) hybrid narrowly leads (CFS 0.827 vs 0.807).

Metric	BM25-primary	Hybrid+RRF
NLI (entailment)	0.925	0.915
SCR (cosine)	0.433	0.438
KFP (key-fact recall)	0.966	0.966
Combined CFS	0.785	0.782
SCHR@5	0.860	0.750
Faithful (≥ 0.55)	93/100	92/100

supporting-chunk hit rate at depth 5 (SCHR@5 0.86 versus 0.75) and one more faithful case (93 versus 92 of 100 under the CFS$\geq$0.55 cutoff). The narrow hybrid edge on the 3-paper anchor (Combined CFS 0.827 versus 0.807, SCHR@5 0.922) does not survive at scale: fusing a single-pass dense encoder that is itself weak on diverse papers, as the retrieval results above show, adds nothing here, which is why BM25-primary is the production default. This metric measures whether retrieval surfaces support for a known claim, not whether the generated answer is faithful; the latter is Section 5.2.

5.2 Generation Faithfulness

We first report the faithfulness of the system's generated answers on the same 51 queries, using the local model gemma4:12b (Gemma Team 2026) for generation. The mean generation-faithfulness score is 0.937: 93.7% of the atomic claims in the generated answers are entailed by the retrieved context. The mean contradiction rate is 0.0, so no answer atom was classified as contradicting its source. Of 269 inline citations checked, 248 are fully supported, 21 partially supported, and none unsupported, a citation-support rate of 0.922. These scores measure grounding to the retrieved context and the absence of outright contradiction, both of which run naturally high for a well-behaved retrieval-augmented system, so we read them conservatively. The zero unsupported-citation count is automated evidence for the indirect citation mechanism, not a claim of perfect correctness against ground truth; assessing that is the job of the human evaluation (Section 6).

Table 4: The same ScholAR pipeline with only the generation model changed, on the 100 diverse cases. Faith. is mean generation faithfulness and Hall. the mean hallucination rate; Cites is the number of inline citations the model emitted and Supp. the fraction of them supported by the cited evidence. Faithfulness varies widely across models; citation support does not.

Model	Faith.	Hall.	Cites	Supp.
gemma4:12b	0.951	0.001	366	0.907
qwen3.5:9b	0.908	0.003	587	0.888
mistral:7b	0.809	0.006	228	0.868
llama3.1:8b	0.719	0.041	81	0.951

5.3 Does the Grounding Hold Across Models?

ScholAR's premise is that grounding lives in the retrieval and application layer, not in the model, so we swapped only the generation model and re-ran the pipeline over the 100 diverse cases (Table 4). Only half of that premise survives. Generation faithfulness does *not* hold constant: it ranges from 0.719 (llama3.1:8b) to 0.951 (gemma4:12b), and llama3.1:8b carries the only non-trivial hallucination rate (0.041). Citation support, by contrast, stays within a narrow band for every model (0.868 to 0.951). The layer the design actually controls, the evidence-identifier citation mechanism, is the stable one; the prose around the citations tracks the strength of the underlying model.

llama3.1:8b is the instructive case: it emits the fewest citations (81, against 587 for qwen3.5:9b) yet has the highest support rate (0.951). It cites well but answers tersely, leaving more prose uncited, which drags its faithfulness down; a high support rate alone is therefore not evidence of a faithful answer, which is why we report both. This refines the model-agnostic claim rather than confirming it: the citation mechanism transfers, answer quality still depends on the model.

5.4 Visual Grounding

On the 18-case visual pilot, retrieval routes to the correct figure or table in all 18 cases, never falling back to the caption-only path. On a keyword-overlap answer-quality proxy (not an LLM judge) gemma4:12b scores 0.731 and qwen3.5:9b (Qwen Team 2026) 0.812; a caption-only ablation scores 0.752 against 0.731 for full vision under gemma4:12b. We do not read that as evidence the image is unnecessary: the proxy rewards lexical overlap rather than visual correctness, and many answers are restated in nearby prose. The cleaner test is Section 5.5, where reading the figure is worth 2.6$\times$ on MRR.

5.5 Multi-Document Localization

We evaluate cross-document localization on M3SciQA (Li et al. 2024), whose *locality* task is exactly ours: given an anchor paper and a question about one of its figures, rank the anchor's bibliography (a mean of 47.9 candidates) to find the reference that answers it. Their 297 gold-labeled locality cases and metric place us against their published baselines (Table 5).

Table 5: Multi-document localization on M3SciQA's locality task (297 labeled cases, mean 47.9 candidate references). Rows marked † are figures published by M3SciQA (Li et al. 2024); the rest are measured here. Resolving the figure with a local multimodal model, rather than retrieving from the question alone, is what moves this task.

System	MRR	R@5
Human expert†	0.796	–
GPT-4o† (cloud)	0.500	–
ScholAR + vision (qwen3.5)	**0.474**	**0.606**
ScholAR + vision (gemma4)	0.455	0.572
GPT-4V† (cloud)	0.400	–
text-emb-3-large† (cloud)	0.297	–
ScholAR BM25 (text only)	0.180	0.242
Best open-source LMM†	0.144	–
BM25† / Random†	0.127 / 0.126	–

Text-only retrieval is barely above chance. BM25-primary reaches MRR 0.180 against an empirical random floor of 0.121 on the pools we rank, and dense MiniLM (0.125) is indistinguishable from random; M3SciQA report 0.127 for BM25 against a floor of 0.126, so our setup reproduces the shape of theirs. This also explains our own 18-case benchmark, where BM25 scored 0.183: the question names an entity that appears only inside a figure, so a signal over titles and abstracts has almost nothing to match against.

The bottleneck is therefore not retrieval but resolving what the question points at. Routing the anchor figure through the same local multimodal model first, then retrieving with the resolved entity, lifts MRR to 0.474 (qwen3.5:9b) and 0.455 (gemma4:12b): 2.6 times the text-only pipeline and 3.3 times the strongest open-source multimodal baseline they report (0.144), ahead of GPT-4V (0.400), and within 0.026 of GPT-4o (0.500), on a laptop. Expert humans remain far ahead at 0.796, so the task is not solved. One caveat keeps this honest: our pipeline *decomposes* the task (vision resolves the entity, then BM25 ranks the bibliography) whereas M3SciQA prompt their baselines to rank papers directly, so this compares systems on a shared task and metric, not under one identical protocol.

5.6 Comparison with Local Baselines

To place ScholAR against alternatives under its own local constraint, we reimplement three baselines on the same model (qwen3.5:9b) and the same cases (51 labeled and 40 diverse): long-context PDF-chat, which puts the whole paper in the model's context; vanilla RAG, with fixed cross-page chunks and dense retrieval; and a local reimplementation of the rerank-and-summarize step of PaperQA2 (Skarlinski et al. 2024). Table 6 shows ScholAR is strongest on the two axes that matter most, generation faithfulness (0.892) and answer correctness (0.563), narrowly ahead of the PaperQA2-style system and well ahead of long-context PDF-chat, which is worst at answering correctly (0.267): retrieval beats putting the whole document in context. ScholAR is not best on every axis, since the terser baselines reach higher citation precision by citing less, but it leads where grounding and correctness are concerned, with a simpler pipeline than the rerank-and-summarize baseline and page citations that are valid by construction. We compare against local reimplementations, not the hosted OpenScholar, PaperQA2, or SciRAG, which depend on cloud models and large datastores; this is a resource-matched local comparison, not a claim to frontier-scale quality.

Table 6: ScholAR versus local baselines on the shared model (qwen3.5:9b) and cases (51 labeled + 40 diverse). Faith. is generation faithfulness; Correct. is must-include recall; Cite-F1 is ALCE-style citation F1. Every emitted page citation is valid (invalid-page rate 0 for all systems).

System	Faith.	Correct.	Cite-F1
PDF-chat (long-context)	0.801	0.267	0.742
Vanilla RAG	0.860	0.340	0.802
PaperQA2-style (RCS, local)	0.872	0.550	0.782
ScholAR	0.892	0.563	0.760

Table 7: Per-model behavior of the unmodified ScholAR pipeline: only the generation model changes. Abstain / Fabricate are measured on the 20 provably-unanswerable questions (Abstain is the fraction correctly declined). Mem. is the loaded footprint in GB (weights plus KV cache) and latency is end-to-end in seconds (mean / p95) over 20 questions on the evaluation laptop; retrieval adds about 40 ms for every model.

	Unanswerable		Cost per query		
Model	Abst.	Fabr.	Mem.	Tok/s	Lat.
qwen3.5:9b	1.00	0.00	6.1	21.9	11.4/14.6
gemma4:12b	0.95	0.05	8.1	16.3	8.8/14.2
llama3.1:8b	1.00	0.00	6.9	26.9	6.3/17.8
mistral:7b	0.90	0.10	6.5	24.9	6.1/8.4

5.7 Abstention on Unanswerable Questions

A grounded system should decline when the answer is not present rather than invent one. Every case above is answerable, so we built 20 provably-unanswerable questions: each paper-specific question is posed against a different paper in which the queried fact is provably absent (verified by exact substring), so the correct response is to abstain. All four models decline on the large majority (Table 7): qwen3.5:9b and llama3.1:8b on every question, gemma4:12b on 19 of 20, mistral:7b on 18 of 20. The few non-abstentions are a benign overlap rather than invention: on a minibatch-size question the wrongly paired paper independently reports a minibatch size, so the model surfaces a real but off-target value. The set is small and synthetic and abstention is scored by a refusal detector we validated by reading every output, so we read this as evidence that grounding induces abstention, not as a precise rate.

Table 8: Human-evaluation results (in progress). Values are placeholders pending expert scoring; they are not estimated. Dimensions are mean 1–5.

Model	Rel.	Cov.	Faith.	Use.	Cite-F1
qwen3.5:9b	–	–	–	–	–
gemma4:12b	–	–	–	–	–
llama3.1:8b	–	–	–	–	–
mistral:7b	–	–	–	–	–

5.8 Local Deployability

We also measured what a query costs on the target machine, timing the answering path (BM25 retrieval, then generation) over 20 questions per model on the same laptop (Apple Silicon, 18 GB). Retrieval is not the bottleneck: BM25 over one paper's chunks returns in about 40 ms regardless of model, so latency is set almost entirely by generation. Every model fits the 18 GB budget (6 to 8 GB loaded, weights plus KV cache) and answers in single-digit to low-double-digit seconds (Table 7), the 7 to 8 billion parameter text-only models fastest at about 6 seconds. No cloud call and no GPU cluster: one laptop runs the loop.

6 Human Evaluation (In Progress)

Automated metrics measure component behavior, not whether a reader would judge an answer correct and well-cited. Following the practice of comparable systems (Asai et al. 2024; Ding et al. 2025; Skarlinski et al. 2024), we designed a model-agnostic human evaluation over the 100-case diverse benchmark (Section 5). Each question is answered by the four local models (qwen3.5:9b, gemma4:12b, llama3.1:8b (Llama Team 2024), mistral:7b (Jiang et al. 2023)) running the *same* pipeline, so only the generation model changes. Blind and in randomized order, expert evaluators score every answer on four anchored 1–5 dimensions (Relevance, Coverage, Faithfulness, Usefulness) and grade each inline citation Supported / Partial / Unsupported. From these we compute per-model means, citation precision, recall and F1, a Friedman test of whether grounding is model-agnostic, and the correlation between human faithfulness and our automated metric. All 350 answers are generated; expert scoring is the remaining step, so Table 8 reports placeholders, not estimates.

We also report, once available, the Friedman-test result on the Faithfulness dimension (to be filled) and the human-to-automated Faithfulness correlation (to be filled), which together test whether ScholAR's grounding holds across local models and whether the automated metric tracks expert judgment.

7 Discussion and Limitations

ScholAR's contribution is a fully local integration whose two genuinely novel elements, page-preserving chunking and indirect citation grounding, are simple and auditable, and in which the generated answers are measurably grounded, with no contradictions and no unsupported citations on the studied set. The limits of that evidence set the agenda for the rest of this section.

The retrieval and retrieval-support benchmarks span 100 cases across 25 papers, but their gold chunks are auto-derived from mined facts rather than hand-labeled, and the mined queries carry lexical overlap that likely flatters lexical retrieval; the 3-paper hand-labeled sets remain the higher-precision anchor. The visual benchmark (18 cases) is small and hand-labeled, so its deltas fall within noise and carry no significance testing, and the abstention probe is likewise small and synthetic (20 auto-constructed negatives scored by a validated refusal detector), so its near-total rates indicate a tendency, not a guarantee. The generation-faithfulness metric rewards grounding to retrieved context and the absence of contradiction, leaving it insensitive to subtle unfaithfulness; closing that gap is what the human evaluation exists for.

Our comparison (Section 5.6) is against local reimplementations of PDF-chat, vanilla RAG, and PaperQA2's rerank-and-summarize step, not against the hosted OpenScholar, PaperQA2, or SciRAG. Those systems depend on cloud-hosted models and large datastores, so a head-to-head would require either a cloud run, contradicting our local-only design, or crippling them to an untested backend; the honest scope of our result is therefore resource-matched local systems, not frontier-scale quality. Finally, cross-document localization is competitive only once the anchor figure is resolved by the vision model; from the question text alone it sits near chance, and even our best configuration (MRR 0.474) remains far below the 0.796 of expert humans on the same task, so the cross-document setting stays the weakest part of the system.

8 Conclusion

We presented ScholAR, a fully local retrieval-augmented system for grounded scientific document comprehension combining page-preserving chunking with indirect citation grounding, extended to visual and multi-document questions. Its generated answers are grounded with a mean score of 0.937 and 92% fully supported citations, none unsupported; retrieval reaches Recall@5 of 0.93 across 25 papers; it leads local baselines on correctness and faithfulness; and on unanswerable questions the models decline rather than fabricate 90 to 100% of the time. Swapping the generation model shows the citation layer transfers while answer faithfulness does not, so grounding is model-agnostic only where the design enforces it. On M3SciQA, resolving the anchor figure locally lifts cross-document localization from near-chance to MRR 0.474, ahead of every open-source baseline reported there and close to GPT-4o, though still far below expert humans. We contribute a 100-case, 25-paper benchmark and a human-evaluation protocol whose scoring will replace the placeholders reported here.

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